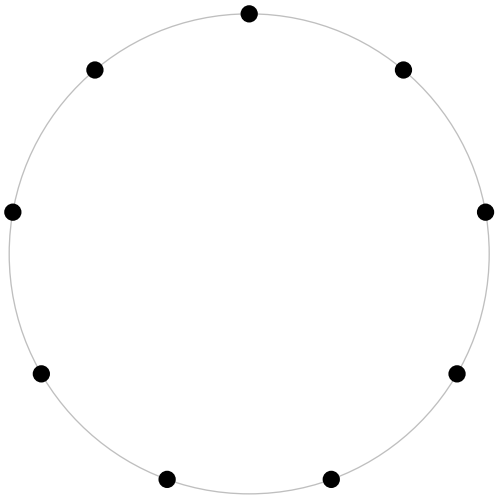
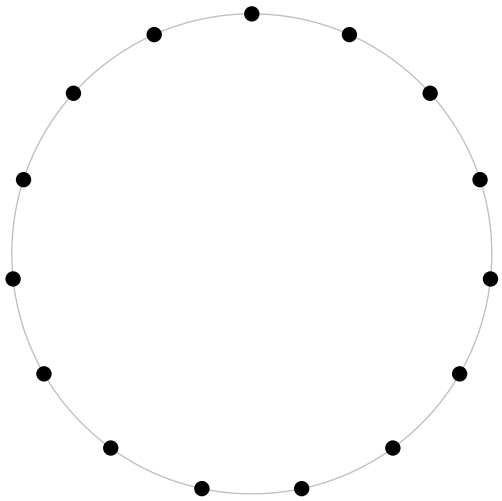
Problem 6: Before you draw each of these, guess how many times you will start and write your guess. Then draw it and write how many times you did start.



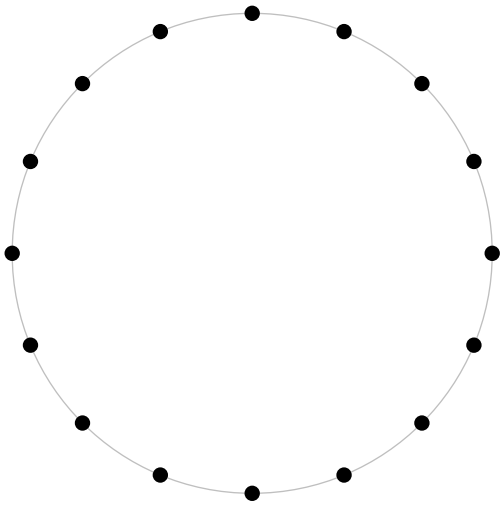
9 dots, hop 2

guess ☐ started ☐



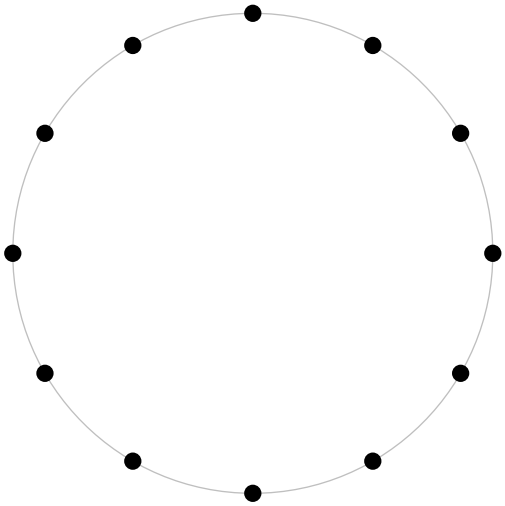
15 dots, hop 5

guess ☐ started ☐



16 dots, hop 6

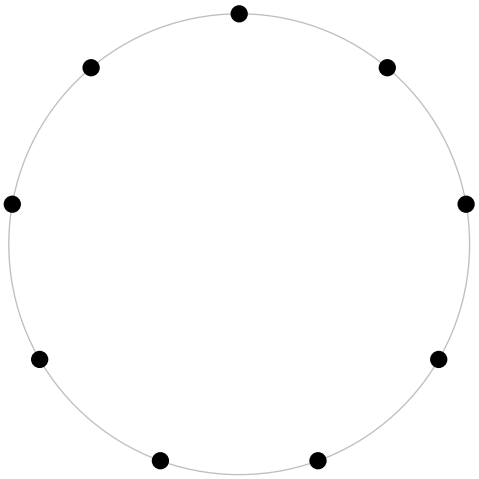
guess ☐ started ☐



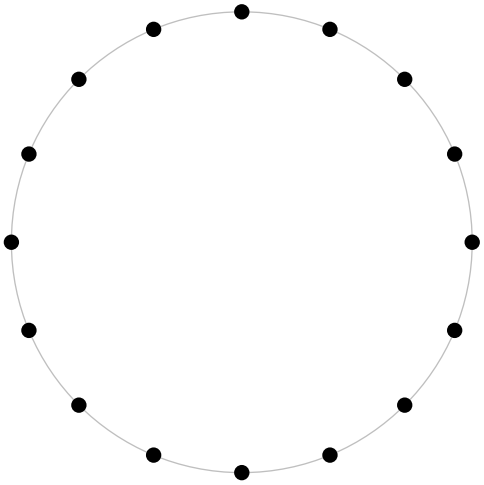
12 dots, hop 8

guess ☐ started ☐

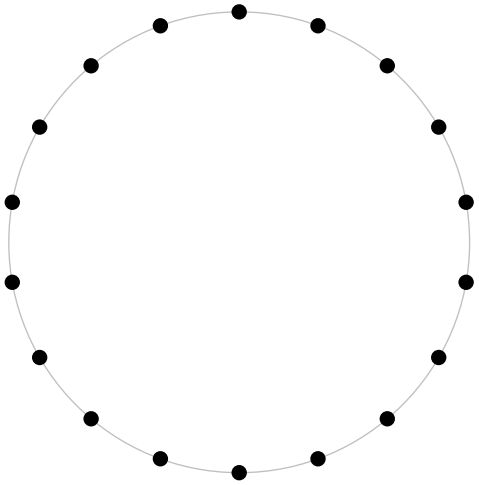
Problem 7: On each of these rings, find a hop bigger than 3 that makes you start exactly 3 times, and draw it. If a ring has no such hop, explain why.



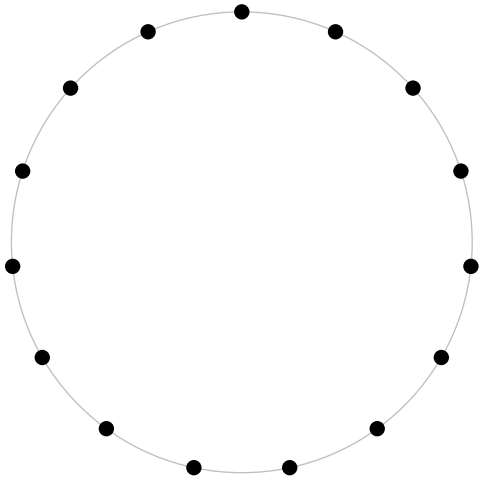
9 dots, hop _____



16 dots, hop _____



18 dots, hop _____



15 dots, hop _____

Problem 8: These messages were put into code with settings nobody wrote down. Find the messages.

H Z A H Y J H U O H C L A L U

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _

W V P U A Z

_ _ _ _ _ _

C U U J C U R O J X U R Y W

_ _ _ _ _ _ _ _ _ _ _ _ _ _

J H U U

_ _ _ _

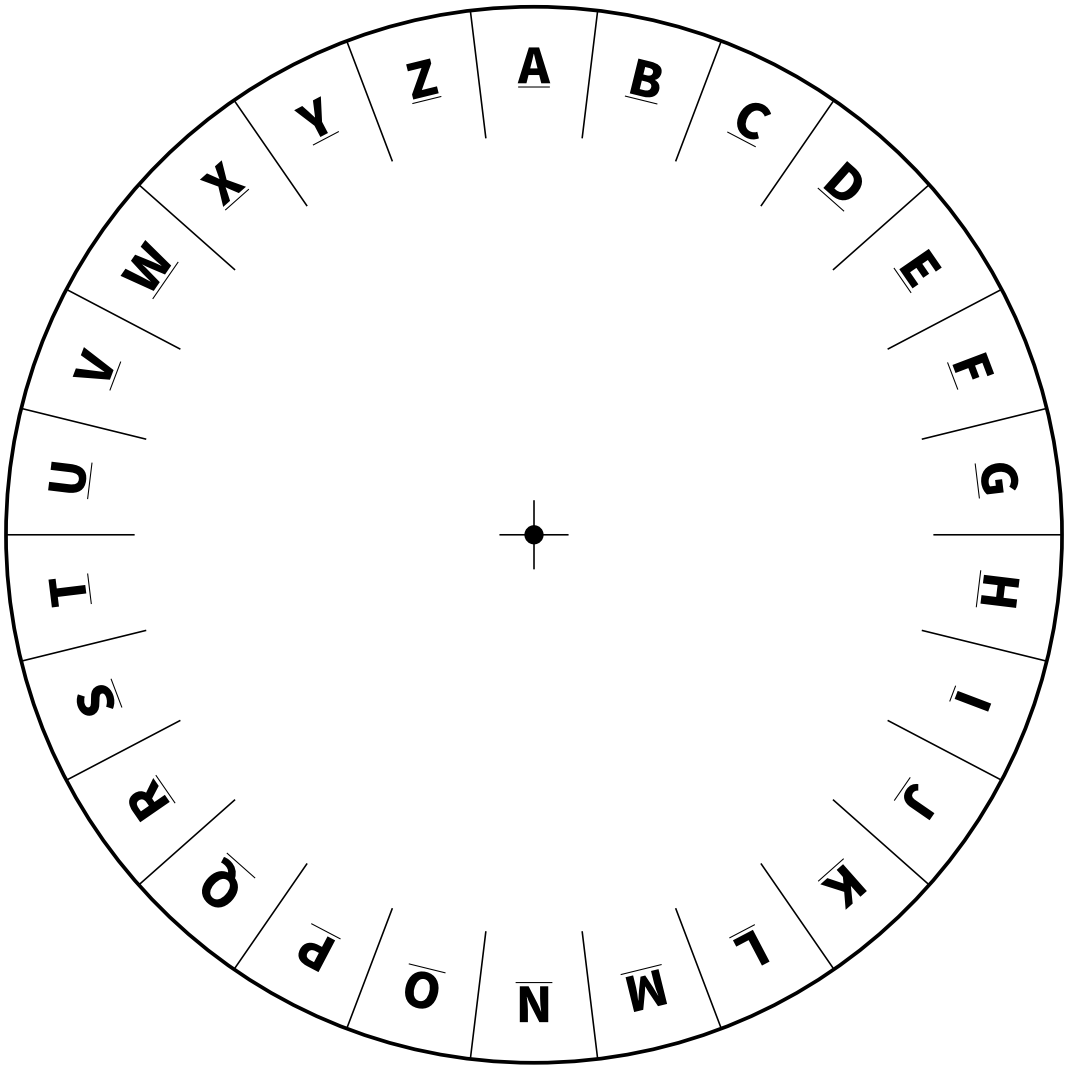
Problem 9: Sam put a word into code. Then he put that code into code again with a second setting, and he got his word back. Find the second setting for each of these first settings.

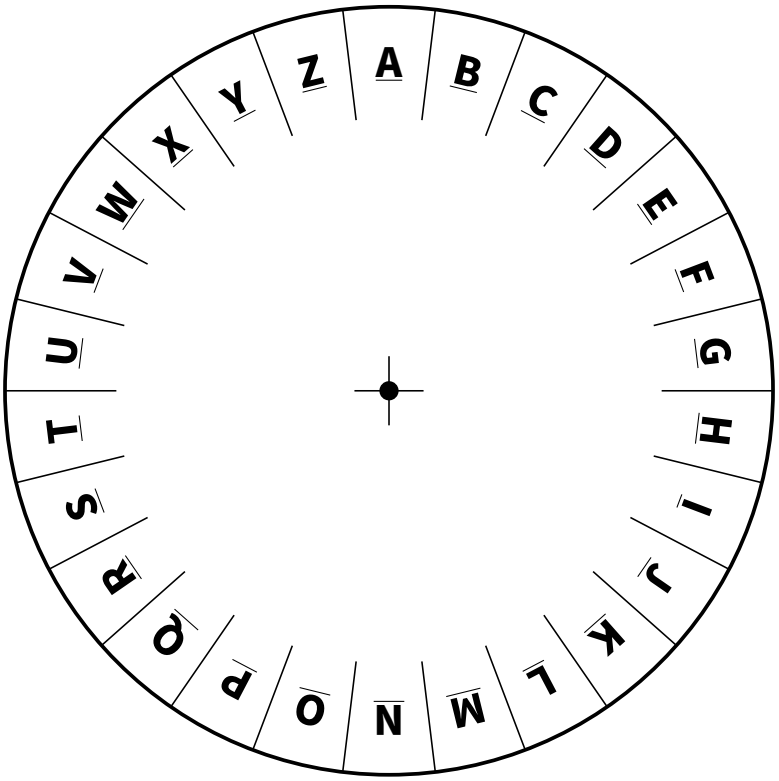
first setting D second setting ☐

first setting H second setting ☐

first setting N second setting ☐

first setting W second setting ☐


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z